Ishaan Maheshwari

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B.Tech Computer Science & Engineering | AI Engineering

SUMMARY

Second-year Computer Science undergraduate focused on applied AI engineering, with hands-on experience building RAG and agentic AI systems. Built core components from first principles before migrating to framework-based tooling, then developed evaluation, prompt-injection testing, and API infrastructure around the system. Focused on building reliable LLM applications and evaluating their behavior through systematic testing.

EDUCATION

B.Tech, Computer Science and Engineering | ABES Engineering College | Expected May 2029
CGPA: 9.3 / 10

PROJECTS

RAG & Agentic AI System | *Independent Project*

- Built an end-to-end RAG and tool-calling agent system from first principles using SentenceTransformers and FAISS for embedding-based retrieval, before migrating the architecture to LangChain.
- Migrated the modular RAG and tool-calling architecture to LangChain, preserving separate retrieval, generation, and tool components.
- Built a custom RAG evaluation pipeline using a gold-question dataset and LLM-as-judge scoring for answer correctness, grounding, hallucination, completeness, and regression testing across iterations.
- Implemented and tested a prompt-injection guardrail, comparing model behavior before and after mitigation and documenting the observed failure mode.
- Exposed the system through a FastAPI REST API with structured request logging for debugging and observability.

Conversation Pruner | *Python • Groq API*

- Built a CLI chatbot that allows selective removal of individual user and assistant messages from the conversation context sent to an LLM, enabling targeted context pruning without restarting the session.
- Implemented index-based pruning with reverse-order deletion to avoid index-shift errors, plus a one-level undo mechanism using a saved history copy.
- Added timestamp-aware conversation handling that detects multi-hour gaps and injects temporary context indicating that the user's context may have changed, without permanently storing the notice.

TECHNICAL SKILLS

Languages: Python, C++
AI/ML: RAG, LLMs, embeddings, vector retrieval, prompt engineering, LLM evaluation, tool calling
Frameworks & Libraries: LangChain, Hugging Face Transformers, SentenceTransformers, FAISS, PyTorch
Backend & Tools: FastAPI, REST APIs, Groq API, Git, GitHub